SMART INDIA HACKATHON 2026	PAN-INDIA 3D CADASTRAL SYSTEM	ISO 19152 COMPLIANT

BHU-AADHAAR 3D

Next-Generation 3D Spatial Cadastre, Subsurface Collision Engine & Vertical Land Registry Portal

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1. Executive Summary & Problem Statement

Modern Indian urban centers (e.g. Mumbai MMR, Delhi NCR, Bengaluru, GIFT City) have evolved rapidly into dense, vertically stacked superstructures with deep multi-level subterranean basements, subterranean metro lines, and underground utility corridors. However, existing land administration systems across India (under DILRMP and Bhu-Naksha) remain fundamentally **two-dimensional (2D)**. Traditional cadastres treat land parcels purely as flat polygons (Latitude, Longitude), assuming a single owner owns the land down to the center of the earth and up to the heavens.

This architectural mismatch creates severe systemic risks:

- **Airspace & Floor Ownership Ambiguity:** In a 40-storey high-rise, dozens of distinct owners share the exact same 2D footprint. Traditional cadastres cannot legally demarcate individual volumetric rights.
- **Catastrophic Subsurface Encroachments:** Deep private foundations and underground parking lots often collide with or breach safety buffers of underground metro tunnels, gas conduits, water mains, and optic fiber networks.
- **Title Disputes & Double Mortgaging:** Absence of a unique, immutable vertical parcel identifier allows fraudulent sale or overlapping conveyance of individual floor units.

The Solution: *Bhu-Aadhaar 3D* provides an ISO 19152 compliant 3D cadastral platform featuring real-time volumetric floor slicing, automated 3D subsurface collision detection, a 14-digit + vertical ULPIN standard, cryptographic title certificate issuance with QR verification, and role-based workflows for Citizens, Surveyors, and Sub-Registrars.

2. System Architecture & End-to-End Pipeline

The project is structured around a modular, decoupled architecture consisting of four core mathematical engines, a relational spatial SQLite database, an interactive Deck.gl GIS visualization layer, and role-based operational modules:

Component	Module Path	Primary Responsibility & Algorithms
Spatial Math Engine	`core/spatial_math.py`	Segments total building height and base elevation into discrete volumetric floor blocks. Supports above-ground superstructures and negative subterranean basements.

3D ULPIN Engine	`core/ulpin_logic.py`	Generates 14-digit LGD-compliant base ULPIN + 3-char vertical suffix (e.g. -004 for floor 4, -U02 for basement 2). Provides bidirectional instant parsing.
Collision & Clash Engine	`core/collision_engine.py`	Performs Haversine surface proximity calculation and 1D vertical overlap checks. Identifies CRITICAL_COLLISION and BUFFER_VIOLATION against metro & utility infrastructure.
Digital Certificate Engine	`core/certificate_generator.py`	Computes immutable SHA-256 cryptographic title hash, generates QR code payload with parcel metadata, and builds downloadable official PDF certificates.
PyDeck 3D Map Studio	`frontend/map_engine.py`	Renders 3D extruded ColumnLayer columns and ground ScatterplotLayer footprints. Features 8 metro viewports, custom camera focus, and theme-adaptive basemaps.
UI Components & Theming	`frontend/ui_components.py`	Injects modern glassmorphic styling, provides persistent Light/Dark mode switcher, KPI metrics row, architectural cross-section, and dossier cards.
Database Storage	`database/spatial_records.db`	Relational SQLite storage storing property IDs, LGD geography, 3D coordinates (lat, lon, base_elevation, total_height), zone, owner, and valuation.

3. Deep Dive: Mathematical & Algorithmic Engines

3.1 Volumetric 3D Parcel Slicing (core/spatial_math.py)

Standard land surveys only record physical footprints. Our spatial segmentation algorithm accepts building base elevation (Z_{base}), total height (H), and floor height ($h = 3\text{m}$ default) to generate precise 3D vertical intervals:

- **Superstructure (Above Ground, $Z_{base} \ge 0$):** For floor index $f \in [0, N-1]$, interval is $[Z_{base} + f \cdot h,\; Z_{base} + (f+1) \cdot h]$. Floor number is $f+1$.
- **Subsurface (Basements, $Z_{base} < 0$):** Extends downwards below surface datum $[-(f \cdot h),\; -((f+1) \cdot h)]$. Floor number is recorded as $-(f+1)$ to denote subterranean depth.

3.2 14-Digit Bhu-Aadhaar + 3D Vertical ULPIN Standard (core/ulpin_logic.py)

India's Ministry of Rural Development specifies a 14-digit Unique Land Parcel Identification Number (ULPIN) derived from Local Government Directory (LGD) codes: SS-DD-SSS-VVV-PPPP (State 2 digits, District 2 digits, Sub-District 3 digits, Village/Ward 3 digits, Plot ID 4 digits).

Our 3D ISO 19152 Cadastral Extension: We append a hyphenated 3-character vertical suffix: -FFF for above-ground floors (e.g., -004 for Level 4) and -UXX for underground basements (e.g., -U02 for Basement 2). The bidirectional decoder instantly unpacks raw ULPIN inputs, validates length and integrity, extracts geographic jurisdiction, and flies the camera to the parcel.

3.3 Subsurface 3D Clash & Encroachment Detection (core/collision_engine.py)

To prevent catastrophic strikes between deep foundations and municipal utility corridors, the collision engine computes both geodesic proximity and vertical intersection:

- **Horizontal Ground Distance:** Uses the Haversine equation over Earth radius $R = 6,371,000\text{m}$:
  $$a = \sin^2\left(\frac{\Delta \phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta \lambda}{2}\right), \quad D = 2R \cdot \arctan2\left(\sqrt{a}, \sqrt{1-a}\right)$$
- **Vertical Interval Overlap (ΔZ):**
  $$\Delta Z = \min(Z_{top1}, Z_{top2}) - \max(Z_{base1}, Z_{base2})$$
- **Severity Classification:** If $D \le 2 \times R_{footprint}$ and $\Delta Z > 0$, flagged as **CRITICAL COLLISION (Direct 3D Encroachment)**. If $D \le 2 \times R_{footprint} + S_{buffer}$ and $\Delta Z > -10\text{m}$, flagged as **SAFETY BUFFER VIOLATION**.

3.4 Cryptographic SHA-256 Title Certificate & QR Engine (core/certificate_generator.py)

Generates an immutable cryptographic fingerprint sealing the vertical airspace parcel: SHA-256(ULPIN | Owner | Lat | Lon | Z_Range | Valuation). Generates a scannable JSON QR payload that any citizen or bank official can scan with a standard smartphone camera to verify title legitimacy on-site.

4. Three Role-Based Operational Workflows

To address the diverse stakeholders of the Indian cadastral ecosystem, the platform provides tailored operational personas with 1-click switching:

Persona Mode	Primary User	Features & Workflows Enabled
Citizen / Homebuyer Mode	Citizens, Flat Buyers, Real Estate Investors, Banks	• Instant title verification & RERA status check • Unit/Floor level dropdown with height bounds • Live 3D Bhu-Aadhaar certificate card preview • 1-Click official PDF title certificate download
Government GIS Surveyor Mode	Municipal Engineers, GIS Officers, Urban Planners	• 3D spatial metrics (Gross Volume m^3, Built-up area m^2, FSI/FAR) • Automated subsurface 3D clash & buffer violation audit • Interactive vertical architectural cross-section stack • 3D parcel coordinate ingestion into SQLite database
Sub-Registrar Mode	Revenue Officers, Land Registrars, Legal Authorities	• Individual vertical airspace deed registration • 6% state stamp duty calculator based on cadastral value • Immutable ownership conveyance in SQLite database • Audit trail logging for pan-India transactions

5. Comprehensive UI/UX Evolution & Recent Changes

A major upgrade was executed to transform the prototype into a commercial-grade, multi-theme portal with a dedicated **Light Mode / Dark Mode switch button**:

- **Prominent Theme Switch Button:** Added an interactive switch button (■■ Light Mode / ■ Dark Mode) embedded in the top portal navbar alongside the Bhu-Naksha Sync pill. Toggling updates st.session_state['theme'] with zero page flicker.
- **Universal CSS Theming System:** Implemented a complete dual-palette design token architecture in frontend/ui_components.py covering Streamlit root canvas, typography, metric cards, inputs, dropdowns, expanders, tabs, radio pills, tables, and scrollbars.
- **Theme-Adaptive Custom HTML Blocks:** Updated all custom cards—property dossier, vertical architectural cross-section, clash alerts, parsed ULPIN banner, map legend, and digital certificates—to seamlessly adapt background, text contrast, and border glow.
- **Basemap Aesthetic Synchronization:** Synchronized the 3D PyDeck basemap default to switch between Minimalist Light (in Light Mode) and Dark Matter (Default) (in Dark Mode), with user manual preference persistence.
- **High-DPI 3D Rendering:** Refined Deck.gl 3D extruded columns with high-contrast neon palettes and glowing footprints to ensure readability under both dark and bright lighting environments.

6. Relational Database Schema (SQLite)

The system stores all pan-India cadastral parcels in database/spatial_records.db under table property_parcels:

Column Name	Type	Cadastral Description
`property_id`	`INTEGER PRIMARY KEY`	Unique Cadastral Parcel ID (e.g. 101, 102)
`name`	`TEXT`	Building / Complex Name (e.g. Seawoods Grand Central)
`city / state`	`TEXT`	Municipal city and Indian state name
`state_code / dist_code`	`INTEGER`	LGD State and District codes for 14-digit ULPIN derivation
`lat / lon`	`REAL`	Precise WGS84 GPS decimal coordinates
`base_elevation`	`REAL`	Elevation datum in meters (negative = subterranean)
`total_height`	`REAL`	Vertical structural height in meters (e.g. 50.0m)
`type`	`TEXT`	Commercial, Apartment, Parking, Transit, Utility
`owner`	`TEXT`	Legal title holder / Agency / Developer
`valuation_cr`	`REAL`	Cadastral valuation in Crores INR (for stamp duty)

7. SIH 2026 Impact & Competitive Advantages

Bhu-Aadhaar 3D provides an end-to-end, working implementation addressing the exact needs of modern smart cities and land governance:

- **Fully Functional Math & Slicing:** Not just mockups—every parcel slices dynamically, computes FSI, and calculates real Haversine & Z-overlap collisions.
- **Standards Compliance:** Aligned with the ISO 19152 Land Administration Domain Model (LADM) and India's DILRMP ULPIN specification.
- **Real Utility:** Solves multi-crore utility breach disputes, accelerates homebuyer bank loan approvals, and unlocks vertical deed registration revenues.

- **Production-Ready UX:** Seamless Dark/Light mode, high-contrast typography, scannable QR codes, and instant vector PDF certificates.